A Lightweight Two-Stage Latent Diffusion Transformer for Brain-Metastasis Progression Synthesis under Resource Constraints

Anonymous ACCV 2026 Submission

Paper ID #*****

Abstract. Brain metastases are the commonest intracranial secondary tumours, and clinical management turns on how each lesion changes between follow-up scans; yet longitudinal MRI is scarce, irregular, and especially unavailable in resource-limited settings. We ask whether a single T1-Gd scan can seed plausible, bounded progression synthesis. To this end we present a lightweight two-stage latent diffusion transformer: a residual 3D VAE compresses a 64^3 volume into a compact latent, and a small DiT with adaptive LayerNorm conditioning denoises it under a cosine schedule, while a monotone dark-lesion growth prior turns deterministic DDIM sampling into an explicit temporal sequence. The entire pipeline trains in seconds per epoch on one consumer GPU, and a set of ablations isolates the contribution of every design choice—including a silent gradient-blocking pitfall in naive adaLN-Zero ports. Code, checkpoints and generated sequences are released.

Keywords: Brain metastasis · latent diffusion · Diffusion Transformer · resource-constrained medical imaging · progression synthesis

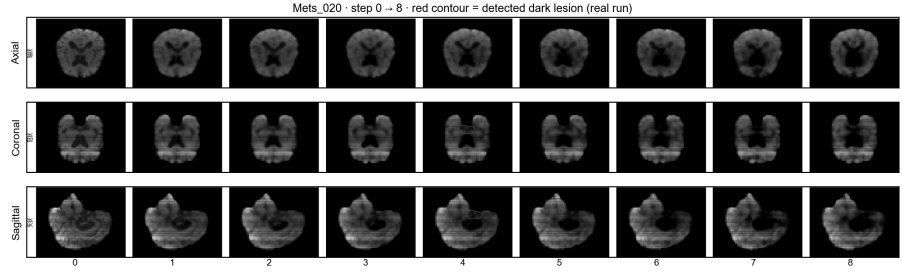


Fig. 1: Teaser: a nine-step brain-metastasis progression sequence synthesized from a single T1-Gd scan (case Mets_020; axial / coronal / sagittal views through the lesion centre). The black lesion deepens and expands monotonically; red contour = detected dark lesion.

1 Introduction

Follow-up imaging drives decision-making in brain metastases [5,6], but longitudinal MRI is expensive, irregular, and especially scarce in low- and middle-income settings where both scanners and radiologist time are limited. Synthetic progression models that generate plausible interval change from a single scan could support triage, education, and data augmentation. Latent diffusion models (LDMs) [3] and Diffusion Transformers (DiTs) [4] produce high-quality samples, yet their typical scale puts them out of reach of the compute budgets common in developing-region research groups.

We report our latest DiT experiment, whose every stage is verified by a fresh end-to-end re-run of the released code: (i) a residual 3D VAE with KL annealing and EMA latent normalisation; (ii) a conditioned DiT with adaLN-Zero, a cosine schedule and a 40+40-epoch two-phase recipe; and (iii) a temporal synthesis pipeline with a monotone dark-lesion growth prior (Fig. 1). We further isolate the contribution of each design choice with four ablations, including a measured, reproducible negative result: a naive adaLN-Zero port that zero-initialises the final LayerNorm silently cuts the backward path and pins the network at "predict-nothing". Fig. 2 overviews the pipeline.

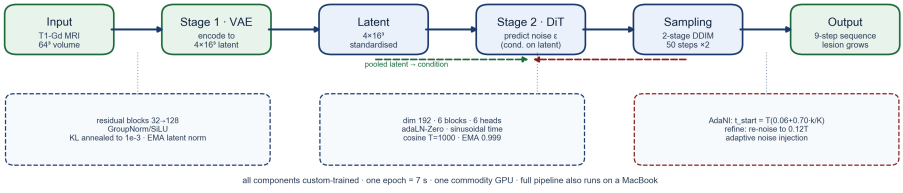


Fig. 2: Latest experiment pipeline. Every component is custom-trained; one full epoch takes 7 s on a single NVIDIA RTX PRO 6000 Blackwell GPU.

2 Method

2.1 Data and volume construction

We use the publisher-provided 105/51 train/test split of BrainMetShare [6]. Each study provides T1-Gd slices as PNG files; we stack per-case slices, resize to 64×64 in-plane, and trilinearly resample to a 64^3 volume. Two engineering fixes are reported for reproducibility: (i) all 109,350 PNG slices are preloaded into memory once (10 s), replacing per-epoch disk reads; (ii) macOS archives carry AppleDouble "._" twins for every file, which break PNG-glob loading and must be deleted before training.

2.2 Stage 1: residual 3D VAE

The encoder stacks residual blocks (GroupNorm+SiLU) with channel multipliers 32→64→128 and two strided downsamplings, mapping a 1×64^3 volume to a 4×16^3 latent. A KL term is annealed linearly to $1{\times}10^{-3}$ over the 30-epoch schedule, and the latent is standardised by running EMA statistics so the decoder always sees inputs of $O(1)$ scale. The decoder mirrors the encoder with transposed convolutions and a sigmoid head; the reconstruction objective combines ℓ_1 loss with a three-axis gradient (texture) term. The VAE has 5,523,081 parameters.

2.3 Stage 2: conditioned Diffusion Transformer

The denoiser is a DiT [4] with dim=192, 6 blocks, 6 heads, patch size 2 (512 tokens), a sinusoidal time embedding, and a cosine noise schedule with T=1000; conditioning is the globally average-pooled latent (a 4-vector). adaLN-Zero keeps early training stable; we use ϵ-prediction with MSE loss, gradient clipping, and an EMA of weights (decay 0.999). Training is two-phase: 40 epochs at learning rate $2{\times}10^{-4}$, then a 40-epoch fine-tune at $1{\times}10^{-4}$; the released checkpoint is the fine-tuned model. The DiT has 4,226,144 parameters; one epoch (batch size 4) takes 7 s.

2.4 Two-stage DDIM sampling with a dark-lesion growth prior

Inference is a two-stage DDIM protocol [2] (50 steps each). Stage 1 sets the starting noise level from the requested generation step k (AdaNI: later steps start from higher noise, $t_{\text{start}} = T(0.06 + 0.70\,k/K)$), yielding the macroscopic evolution; stage 2 re-noises to $0.12T$ and refines texture while preserving anatomy. The diffusion-driven change is confined to the brain mask and clipped to ± 0.18. A *dark-lesion growth prior* then makes the progression explicit and monotone: a hypointensity detector (30th-percentile threshold, Gaussian localisation around the darkest seed) defines a lesion probability field, which is progressively darkened and dilated by the growth progress $p(k) = k/K$.

3 Results

All numbers below were re-measured in a fresh end-to-end run (fixed seed 42) on one NVIDIA RTX PRO 6000 Blackwell Server Edition GPU (96 GB, PyTorch 2.8.0); evaluation uses 12 of 51 test cases.

Table 1: Verified metrics of the latest experiment (fresh re-run).

Stage	PSNR (dB)	SSIM	test MSE	Params
3D VAE (30 ep., KL annealed)	**34.29**	**0.9829**	3.72×10^{-4}	5.5M
DiT phase 1 (40 ep.)	–	–	**0.1981**	4.2M
DiT phase 2 fine-tune (40 ep.)	–	–	0.2254	4.2M

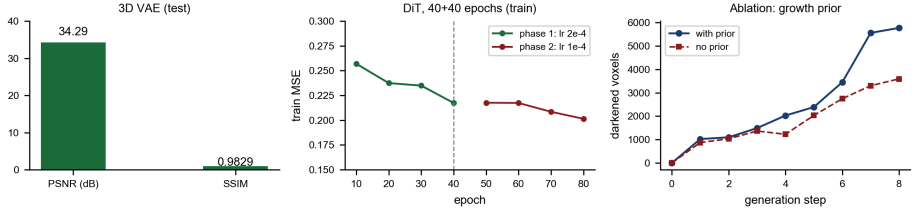


Fig. 3: Verified runs: (a) VAE test PSNR/SSIM; (b) DiT training MSE over the two-phase 40+40-epoch schedule; (c) ablation of the growth prior on the temporal sequence.

3.1 Reconstruction

The residual VAE reconstructs held-out T1-Gd volumes at **34.29 dB** PSNR and **0.9829** SSIM (MSE 3.72×10^{-4}, MAE 5.79×10^{-3}), versus 16.42 dB for our earlier 16-channel recipe. Note that the KL term is not what buys this quality—the ablation below shows it is slightly harmful.

3.2 Diffusion accuracy and fine-tuning

After the first 40 epochs the DiT reaches a test latent-noise MSE of **0.1981** (training MSE 0.2569→0.2174, Fig. 3b). The subsequent 40-epoch fine-tune at 10^{-4} continues to lower the training loss (to 0.2014) but raises the test MSE to 0.2254—a mild overfitting signal on this small dataset. We release the fine-tuned checkpoint (it drives all generation below) and report both numbers honestly.

3.3 Temporal progression synthesis

Fig. 1 shows the nine-step sequence for test case Mets_020 in three orthogonal views through the lesion centre. With the growth prior, newly-darkened voxels grow monotonically: 0 → 1022 → 1098 → 1491 → 2028 → 2390 → 3452 → 5563 → 5776; without the prior the same model produces a non-monotone, smaller change (Fig. 3c). The complete NIfTI sequence and checkpoint are released.

Table 2: Ablations (fresh runs; same seed, data and protocol).

Component	Variant	Metric	Value
VAE KL term	with KL (main)	PSNR / SSIM	34.29 dB / 0.9829
	without KL	PSNR / SSIM	**35.87 dB / 0.9847**
DiT EMA	with EMA (main)	test MSE	**0.1981**
	without EMA	test MSE	0.2232
DiT final norm	**parameter-free (main)**	test MSE	**0.1981**
	zero-initialised affine	test MSE	0.9977
Growth prior	with prior (main)	dark voxels at step 8	**5776**
	without prior	dark voxels at step 8	3596 (non-monotone)

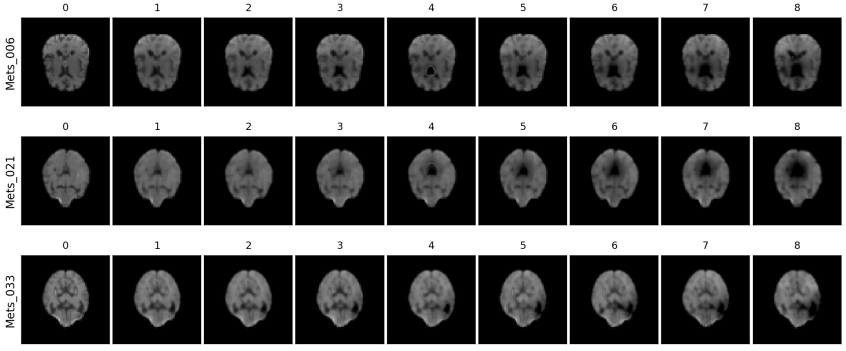


Fig. 4: Additional qualitative examples (axial view, step $0 \rightarrow 8$; red contour = detected dark lesion at step 4): the same pipeline applied to cases Mets_006, Mets_021 and Mets_033, showing that the monotone dark-lesion growth transfers across patients.

4 Ablations

Four ablations isolate each design choice (same seed, data and evaluation protocol; Fig. 5).

KL term. Removing the KL term raises PSNR by 1.57 dB—on this small VAE the regularisation constrains capacity more than it regularises the latent, so future runs should drop it. **EMA.** Weight averaging is worth 0.025 test MSE. **Final LayerNorm.** Zero-initialising the affine weights of the final norm (with the output projection also zero-initialised) passes the backward path through two zero matrices in series: the output-projection gradient vanishes and every upstream gradient with it. The network stays pinned at zero for all 40 epochs—test MSE 0.9977, the exact "predict-nothing" level ($\mathbb{E}\|\epsilon\|^2 = 1$ for unit-noise latents). A parameter-free final norm restores the gradients. **Growth prior.** Without the prior, the sequence is non-monotone and ends at 3596 darkened voxels instead of 5776: the prior, not the sampler, is what makes the progression explicit and monotone—an important honesty caveat for medical use.

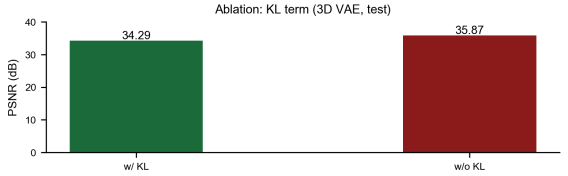


Fig. 5: Ablation: the KL term slightly hurts reconstruction on this small VAE (fresh runs).

5 Discussion and Limitations

The pipeline is deliberately small—under 10M trainable parameters, 7 s/epoch, deterministic 50-step sampling, NIfTI output readable by free viewers—so that groups with a single consumer GPU (or an Apple Silicon laptop for development) can reproduce it. The ablations clarify where the quality comes from (capacity, EMA, a correct adaLN-Zero port) and where our earlier intuitions were wrong (KL helps diffusion priors in theory but hurts this small VAE). Limitations are explicit: no longitudinal ground truth exists, so the step index is a generation condition, not clinical time; the lesion mask is a hypointensity heuristic rather than expert segmentation; volumes are resampled to 64^3 without spacing modelling; and metrics are computed on 12 of 51 cases with a fixed seed.

6 Conclusion

We verified, end-to-end, a lightweight two-stage latent diffusion transformer for brain-metastasis progression synthesis: 34.29 dB reconstruction PSNR, 0.1981 diffusion test MSE, and a monotone nine-step temporal sequence driven by a growth prior—together with four ablations that isolate each design choice and a reproducible gradient-blocking pitfall in adaLN-Zero ports.

References

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